

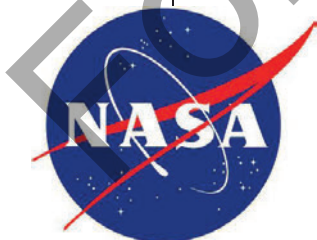
Balloon Program Office



Statement Of Work

Fabrication of Mobile Launch Vehicle

Contract No: 80GSFC22CA019

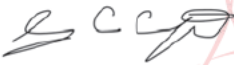


National Aeronautics and
Space Administration

Goddard Space Flight Center
Wallops Flight Facility
Wallops Island, Virginia

Statement Of Work
Fabrication of Mobile Launch Vehicle
Contract No: 80GSFC22CA019

Signature/Approval Page

	Name	Signature	Date
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Approved	Amy Canfield 820 / Contracting Officer's Representative		Digitally signed by Amy Canfield Date: 2025.05.29 10:29:37 -04'00'

Revision History		
Revision	Effective Date	Description of Changes
0	April 24, 2025	Initial Release
1	May 2025	Added in schedule C for the fabrication. Minor edits in schedule A and B denoted by underlines.

BACKGROUND

National Aeronautics and Space Administration's (NASA) Balloon Program Office (BPO), [REDACTED], utilizes specially fabricated Mobile Launch Vehicles (MLVs), designed to facilitate the launch of scientific balloon payloads. The units currently in use are beginning to show their age. BPO [REDACTED] has put a call to vendors to fabricate new, updated MLVs based heavily on the design of the existing launch vehicle housed at the Columbia Scientific Balloon Facility's (CSBF) facility in Palestine, TX. Unlike the existing MLV, the New MLV design will utilize a separate MLV chassis and powerplant. The Caterpillar Model 966M wheel loader was selected [REDACTED] to serve as the MLV powerplant [REDACTED] and two (2) units were procured. The MLV Chassis design was completed under [REDACTED] 06. BPO is now seeking to procure the fabrication of the required MLV Chassis.

SCOPE

This Statement of Work (SOW) describes the tasks associated with the fabrication, and acceptance testing of a new Mobile Launch Vehicle (MLV) to be operated and managed [REDACTED] under the NASA Sub-Orbital and Special Orbital Projects Directorate.

SCHEDULE A – FABRICATION VENDOR(S) VERIFICATION OF CAPABILITIES**SCHEDULE B – MLV DESIGN DOCUMENT EXCHANGE AND FABRICATION PLANNING****SCHEDULE C – MLV FABRICATION****TRAVEL REQUIREMENTS**

BPO [REDACTED] personnel may be required to travel to various vendor locations to support the fabrication of the MLV chassis. Furthermore, [REDACTED] personnel may be required to transport one or both of the Caterpillar 966M wheel loaders to various vendor locations to facilitate fabrication, and/or acceptance test of the MLV chassis.

QUALIFICATIONS AND GENERAL REQUIREMENTS

Task documentation and deliverables as stated in this SOW and as set forth under [REDACTED]

[REDACTED] GSFC 52.235-92 Reports of Work (CORE/IDIQ) (Aug 2013).

“820-REQ-13 MLV Chassis” will serve as the MLV requirements document under this Statement of Work.

STANDARDS, SPECIFICATIONS, AND/OR NASA REQUIREMENT DOCUMENTS

The following standards, specifications, and/or NASA requirement documents are applicable to all schedules of this task order.

- OF-603-02-P LECC Document (ref 08-20010-05)
- OM-300-200-00-SW.A Launch Head Assembly (aka Crane Head Assembly)
- NASA Systems Engineering Handbook (ref 08-20010-WI-07)
- NASA-STD-5005D (ref in 08-20010-DOC-01)
- NASA-STD-5008B (ref in 08-20010-DOC-01)
- NFPA 780 (ref in 08-20010-DOC-01)
- GPR 1820.1C: section 1.9c (ref in 08-20010-DOC-01)
- OSHA/ANSI Z359 (ref in 08-20010-DOC-01)
- OSHA 29 CFR 1910 (ref in 08-20010-DOC-01)
 - OSHA 29 CFR 1910.29
 - OSHA 1910.56
 - OSHA 1910.140
 - OSHA 29 CFR 1926.753(e)2 (assembly only)
- Additional documents may be called out on the drawings in the MLV Design Package.

The following documents constitute the MLV Design Package. Access to the STEP files to the drawings will be made available on request.

- 820-REQ-13B MLV Chassis Requirements
- 08-20000 MLV Assembly
- 08-20000 NBOC MLV Documentation and Reports
- 08-20001 Chassis Assembly
- 08-20002 Boom Assembly package
 - 08-20002 Boom, NBOC MLV
 - 08-20002-01 Mechanical Stop, Upper Boom
 - 08-20002-02 Mechanical Stop, Lower Boom
 - 08-20002-03 .25 Wear Block, Lower Boom
 - 08-20002-04 .25 Wear Block, Upper Boom
 - 08-20002-05 .375 Wear Block, Boom
 - 08-20002-06 Block Off Plate, Boom
 - 08-20002-07 Sensor Mounting Plate, Boom
 - 08-20002-08 Sensor Bracket, Boom
 - 08-20002-09 Whisker Mount, Boom
 - 08-20002-100 Lower Boom
 - 08-20002-133 MLV Lifting Point, NBOC, MLV
 - 08-20002-200 Upper Boom, Boom
 - 08-20002-300 Boom Hinge Pin, Boom
 - 08-20002-301 Pin Retaining Plate, 0.25 STK
 - 08-20002-400 Upper Boom Walkway, Boom
 - 08-20002-500 Hydraulic Cylinder Pin
 - 08-20002-600 Upper Boom Transition, Boom
 - 08-20002-601 Port/Starboard Wheel Block, Upper Boom Transition
- 08-20003 Frame Assembly package
 - 08-20003 Frame Assembly
 - 08-20003-01 L-Angle, Frame Assembly

- 08-20003-02 L-Angle, Cut, Frame Assembly
- 08-20003-03 Long L-Angle Frame Assembly
- 08-20003-04 Stair Bracket, Frame Assembly
- 08-20003-05 Axle Hinge, Frame Assembly
- 08-20003-06 Frame Roller, Frame
- 08-20003-07 Bearing Spacer, Frame Assembly
- 08-20003-08 Mounting Bracket, Angular Position Sensor, Frame Assembly
- 08-20003-09 Skirt, Scissor Lift, Frame
- 08-20003-10 Angle Sensor Adapter, Frame
- 08-20003-11 Pillow Block, Frame
- 08-20003-12 Aft Scissor Lift Cover, Frame
- 08-20003-13 Stair Landing Support, Frame
- 08-20003-14 Plate, Hydraulic Controller Mount, Frame Assembly
- 08-20003-15 Frame Roller Pin, Frame
- 08-20003-16 Manifold Shroud, Frame Assembly
- 08-20003-17 1 1/2-16 Threaded Enerpac Housing Block, Frame Assembly
- 08-20003-18 2 3/4-16 Threaded Enerpac Housing Block, Frame Assembly
- 08-20003-19 Enerpac Coupling, Frame Assembly
- 08-20003-20 Hydraulic Diverter Block, Frame Assembly
- 08-20003-100 Frame Rail, Frame
- 08-20003-200 Pillar Weldment, Frame
- 08-20003-300 Axle Housing, Frame
- 08-20003-400 Stair Landing, Frame
- 08-20003-500 Stair Railing, Frame
- 08-20003-600 Walkway Support, Frame
- 08-20004 Launch Deck Assembly package
 - 08-20004 Launch Deck
 - 08-20004-01 Upper Cage Bar, Launch Deck
 - 08-20004-100 Cage, Launch Deck
 - 08-20004-200 Modified Scissor Lift, Launch Deck
 - 08-20004-201 Mounting Rail, Launch Deck
- 08-20005 Generator Set Pallet Assembly package
 - 08-20005-01 High Pressure Sensor Adapter, Gen Set Pallet Assembly
 - 08-20005-051 Hydraulic Power Unit, Gen Set Pallet Assembly
 - 08-20005-052 Generator Pallet Assembly, Gen Set Pallet Assembly
 - 08-20005-053 Gen Set Pallet Assembly, Fuel Tank Mount
 - 08-20005-100 Pallet, Hydraulic Tank and Pump, Gen Set Pallet Assembly
 - 08-20005-200 Hydraulic Tank, Gen Set Pallet Assembly
 - 08-20005-300 Fuel Tank Pallet, Support Equipment
 - 08-20005-400 Generator Pallet, Support Equipment
- 08-20007-200 Cradle, Axle Cart
- 08-20008-01 Document Tree
- 08-20009 Semi Truck, Chassis Interface package
 - 08-20009 Semi/Chassis Interface
 - 08-20009-01 Air-Hydraulic Brake Bracket, Semi/Chassis Interface
 - 08-20009-02 Air-Hydraulic Brake Linkage, Semi/Chassis Interface
 - 08-20009-100 Semi Truck/Chassis Interface
- 08-20010 NBOC MLV Program Documentation and Reports

- 08-20010-DOC-01 Mobile Launch Vehicle Chassis System Requirements Matrix
- 08-20010-WI-07 NBOC MLV Verification and Validation Plan
- 08-20010-DOC-08 Fabrication Recommendation and Procedures Document
- 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan
- 08-20010-DOC-09 Maintenance Documentation
- 08-20010-DOC-02 Mobile Launch Vehicle Chassis Risk Matrix
- 08-20010-DOC-03 Mobile Launch Vehicle Chassis System Safety Analysis
- 08-20000-DOC-01 NBOC MLV Analysis report
- 08-20009-DOC-01 Semi Truck, Chassis Analysis
- 08-20009-DOC-02 Structural Calculation Memo
- 08-20011 Model Hydraulic Cylinder package
 - 08-20011-SCH-01 Hydraulic Schematic, High Flow
 - 08-20011-DOC-01 Airline Hydraulics Functional Specification
- 08-20012 Service Brake System package
 - 08-20012-50-01 Axle Assembly, Port Side
 - 08-20012-50-02 Axle Assembly, Starboard Side
 - 08-20012-01 Service Caliper Bracket, Axle
 - 08-20012-02 Caliper Shield, .125 STK, Axle
 - 08-20012-03 Axle Tension Plate, 0.5 STK, Axle
 - 08-20012-04 Parking Caliper Bracket, Axle
 - 08-20012-05 Axle Shaft, Axle
 - 08-20012-06 Spline Adapter, Axle
 - 08-20012-100 Axle Hinge Pin, Axle
- 08-20013 Boom Failsafe package
 - 08-20013 Boom Failsafe
 - 08-20013-01 Wedge Block, Boom Failsafe
 - 08-20013-02 Stage 2, Boom Failsafe
 - 08-20013-03 Stage 3, Boom Failsafe
 - 08-20013-04 Internal Slide, Boom Failsafe
 - 08-20013-05 Tapered Pin, Boom Failsafe
 - 08-20013-100 Stage 1, Boom Failsafe
 - 08-20013-200 Stage 4, Boom Failsafe
 - 08-20013-300 Spring Pin Plunger, Boom Failsafe
 - 08-20013-400 Lower Spring Pin Body, Boom Failsafe
 - 08-20013-500 Foot, Boom Failsafe
 - 08-20013-600 Upper Spring Pin Body, Boom Failsafe
- 08-20014 Bucket Interface
- 08-20015 Folding Safety Railing package
 - 08-20015-051 FWD Safety Rail, Safety Railing
 - 08-20015-052 FWD Mid Safety Rail, Safety Railing
 - 08-20015-053 AFT Mid Safety Rail, Safety Railing
 - 08-20015-054 AFT Safety Rail, Safety Railing
 - 08-20015-01 Main Top Rail, Safety Railing
 - 08-20015-02 Upright Rail, TS2X2X0.3125 STK, Safety Railing
 - 08-20015-03 Upright Bracket, Safety Railing
 - 08-20015-04 Narrow Midrail Bracket, 0.25 STK, Safety Railing
 - 08-20015-05 AFT Tapered Toeboard, 0.25 STK, Safety Railing
 - 08-20015-06 Toeboard, Safety Railing

- 08-20015-07 Midrail Bracket 2, 0.25 STK, Safety Railing
- 08-20015-08 Toeboard End Plate, 0.25 STK, Safety Railing
- 08-20015-09 Locking Wedge, Safety Railing
- 08-20015-10 Wedge Stop Block, 0.25 STK, Safety Railing
- 08-20015-11 FWD Tapered Toeboard, 0.25 STK, Safety Railing
- 08-20015-12 Upright Top Bracket, Safety Railing
- 08-20015-13 Toeboard Mount, Safety Railing
- 08-20015-14 Flight Ladder Retainers “Whiskers”, Safety Railing
- 08-20015-15 Railing Upright Foot, Safety Railing
- 08-20015-100 FWD Tapered Toprail, Safety Railing
- 08-20015-200 Step Toprail, Safety Railing
- 08-20015-300 Toprail End Cap, Safety Railing
- 08-20015-400 Handrail Transition
- 08-20015-500 “Whisker” Mount, Safety Guardrail
- 08-20015-600 AFT Tapered Toprail, Safety Railing
- 08-20015-700 AFT Handrail Transition
- 08-20015-800 Middle Guardrail Segment
- 08-20016 MLV Wheel Cart
- 08-20017 Chassis Jack Assembly package
 - 08-20017 Chassis Jack Assembly
 - 08-20017-01 Pin Adapter, Chassis Jack
 - 08-20017-02 Cover Plate, .125 STK, Chassis Jack
 - 08-20017-100 Chassis Jack Top, Chassis Jack
 - 08-20017-200 Chassis Jack Body, Chassis Jack
 - 08-20017-300 Jack Foot, Chassis Jack
 - 08-20017-400 Plunger, Chassis Jack
 - 08-20017-500 Top Pin, Chassis Jack
 - 08-20017-600 Bottom Pin, Chassis Jack
- 08-20018 NBOC II MLV Electrical Schematic package
 - 08-20018-10 Roller Switch Assembly
 - 08-20018-DOC-01 NBOC MLV Power and Control ICD
 - 08-20018-SCH-01 NBOC II MLV – Aux Power System
 - 08-20018-SCH-02 NBOC II MLV – Control System
- 08-20019 Staircase, CAT966M Overhead Walkway package
 - 08-20019 Staircase, CAT966M Overhead Walkway
 - 08-20019-01 Staircase Stringer, CAT966M Overhead Walkway
 - 08-20019-02 Hinge Mounting Plate, Overhead Walkway
 - 08-20019-100 AFT Walkway, Overhead Walkway
 - 08-20019-200 Intermediate Boom Extension, Overhead Walkway
 - 08-20019-300 Mounting Leg, Overhead Walkway
 - 08-20019-400 Transition, Overhead Walkway
 - 08-20019-500 CAT966M Overhead Walkway
 - 08-20019-600 Long Rail, Overhead Walkway
 - 08-20019-700 Short Rail, Overhead Walkway
 - 08-20019-800 Lower Stair Support, Overhead Walkway
 - 08-20019-900 Vertical Railing, Safety Railing
- 08-20020 CAT966M Boom Tension Rod package
 - 08-20020 CAT966M Boom Tension Rod

- 08-20020-01 Replacement Pin, tension Rod
- 08-20020-02 Clevis, Tension Rod
- 08-20021 Spring Locking Pin Assembly package
 - 08-20021 Spring Locking Pin Assembly
 - 08-20021-100 Pin Housing, Spring Locking Pin Assembly
 - 08-20021-01 Spring Pin, Spring Locking Pin Assembly
 - 08-20021-02 Captive Plate, Spring Locking Pin Assembly
 - 08-20021-03 Pin Linkage, Spring Locking Pin Assembly
 - 08-20021-04 Hydraulic Linkage, Spring Locking Pin Assembly
 - 08-20021-05 Sensor Trigger, Spring Locking Pin Assembly

For Reference only

SCHEDULE A – FABRICATION VENDOR(S) VERIFICATION OF CAPABILITIES

TASK DESCRIPTION

- This schedule will have the Contractor review capabilities and select vendor(s) for purposes of fabricating a minimum of one (1) and maximum of three (3) MLV Chassis.

MATERIALS, EQUIPMENT, AND/OR TOOLS

- N/A.

STANDARDS, SPECIFICATIONS, AND/OR NASA REQUIREMENT DOCUMENTS

- OSHA 29 CFR 1910: General Industry Standards
- See MLV Design Package in the General Requirements

TASK DELIVERABLES AND MILESTONES

- Presentation of Fabrication vendor(s) selection
 - Electronic delivery to NASA BPO.

REQUIREMENTS

- Contractor shall procure Fabrication vendor(s) that are capable of completing the MLV chassis fabrication in accordance with the MLV Chassis Design Package in the General Requirements.
 - Contractor shall ensure Fabrication vendor(s) compliance in accordance with 820-REQ-13, Sections 3.2.4, 3.2.5, and 3.5
 - Contractor may choose multiple vendors for the fabrication of the MLV.
 - If multiple vendors, then the Contractor shall ensure compliance for each vendor.
 - If multiple vendors, then the Contractor shall also identify the Fabrication vendor(s) responsible for integration and testing of the fabricated MLV
- At vendor selection, the Contractor will document and present to NASA BPO the following:
 - At a minimum, the presentation shall include:
 - The list of interested vendor(s).
 - The criteria used to determine vendor(s) capability of fabricating the MLV chassis.
 - If multiple vendors, then the responsibility of each selected vendor shall be identified.
 - The final ranking of vendor(s) that identifies the chosen vendor with rationale.
 - At a minimum, the NASA BPO COR and NASA Task Manager must be present for such presentation.
 - Contractor shall deliver the presentation to NASA BPO.

SCHEDULE B – MLV DESIGN DOCUMENT EXCHANGE AND FABRICATION

PLANNING

TASK DESCRIPTION

- This schedule is the planning portion of the fabrication. It may be performed in parallel or serially to the fabrication effort in Schedule C.

MATERIALS, EQUIPMENT AND/OR TOOLS

- Government Owned / Government Furnished - Caterpillar 966M wheel loader, if requested.
- Government Furnished Property – MLV Launch Head, if requested.
- Material as noted in drawings.
- Equipment for inspections as noted in drawings.

STANDARDS, SPECIFICATIONS, AND/OR NASA REQUIREMENT DOCUMENTS

- OSHA 29 CFR 1910: General Industry Standards
- NPR 8715.1: NASA Safety and Health Programs
- See MLV Design Package in the General Requirements, including:
 - 820-REQ-13B MLV Chassis Requirements
 - 08-20010-DOC-08 Fabrication Recommendations and Procedures Document
 - 08-20010-DOC-05 NBOC MLV Acceptance Testing Document
 - 08-20010-DOC-09 Maintenance Documentation

TASK DELIVERABLES AND MILESTONES

- Fabrication vendor(s)'s comments, questions, and possible changes to MLV Design Package in the General Requirements
 - Electronic format to NASA TM
- Fabrication vendor(s)'s comments, questions, and possible changes to 08-20010-DOC-08 Fabrication Recommendations and Procedures Document
 - Electronic format to NASA TM
- Fabrication vendor(s)'s comments, questions, and possible changes to 08-20010-DOC-09 Maintenance Documentation.
 - Electronic format to NASA TM
- Updated 08-20010-DOC-09 Maintenance documentation, draft version.
 - Electronic format to NASA BPO COR and TM
- Fabrication vendor(s)'s comments, questions, and possible changes to 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan
 - Electronic format to NASA TM
- Updated 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan
 - Electronic format to NASA BPO COR and TM
- Updated MLV Design package
 - Electronics format to NASA BPO COR and TM.
- NBOC MLV Post Acceptance Test Inspection Plan, draft version.
 - Electronic format to NASA BPO COR and TM
- MLV Fabrication Inspections and Tests Document
 - Electronic format to NASA BPO COR and TM

- Bi-weekly high level status summary
 - Electronic format to NASA BPO COR and TM

REQUIREMENTS

- The MLV Chassis Design is owned by NASA BPO. Access to the STEP Files will be provided on request.
- Contractor shall provide a single point of contact (POC) to serve as the primary technical interface for the MLV project to NASA Task Manager (TM).
 - The Contractor POC shall schedule meetings with NASA, as needed, to provide the progress and status of the project.
 - Meeting attendance shall be by telecon or on-line meeting platform.
 - Contractor POC shall provide bi-weekly high-level status summary to NASA TM and NASA BPO COR.
 - The high-level status summary shall include all support elements of the SOW.
- Contractor shall develop and provide a schedule for the deliverables of Schedule B.
 - The schedule shall include, at a minimum, the following:
 - Timeline for each deliverable.
 - Tentative review dates.
 - The schedule shall be delivered within two (2) weeks of award to Fabrication vendor(s).
- Contractor shall facilitate a review of the MLV Design Package in the General Requirements.
 - At a minimum, the review shall include the Fabrication vendor(s).
- Contractor shall provide consultation services for to the Fabrication vendor(s) for:
 - Potential design changes of the MLV chassis requested during the fabrication phase.
 - The intended operational implementation of the MLV chassis.
 - The end-use operational environments. (820-REQ-13, section 3.6.5)
 - The support of updating trade studies associated with the maintenance and life cycle considerations of the MLV chassis.
 - Development of updates for the 08-20010-DOC-09 maintenance documentation.
 - Updates to 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan.
 - This plan is inclusive and representative of all the requirements levied and is a means of verifying the design, requirements, and capabilities of the system.
 - This plan shall include both the MLV Chassis and the fully integrated MLV.
 - This plan shall include the procedures for each test. This can be within the 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan or as individual documents.
 - Development of a Draft NBOC MLV Post Acceptance Test Inspection Plan.
 - The plan shall:
 - Ensure the operational integrity of the MLV chassis and fully assembled MLV.
 - Include draft procedures for each test in the 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan
 - Identify what will be documented in the NBOC MLV Post Acceptance Test Inspection Report
 - Identify what will be inspected on the fully assembled MLV to ensure its operational integrity.
 - Development of MLV Fabrication Inspections and Tests Document

- The ~~plan~~ plan document shall:
 - Identify inspections and/or tests that will be performed during fabrication.
 - Identify any certifications required to perform the inspections and/or tests.
- Contractor shall provide any other reasonable support required for the successful fabrication of the MLV chassis.
 - Reasonable support may include, but is not limited to:
 - Facilitating access to a Caterpillar 966M in the most cost-effective method possible. This may be through leasing a unit for them to host at their facility, transporting a unit that is currently in the Contractor's possession to the Fabrication vendor(s), and /or hosting the Fabrication vendor(s) at the CSBF facility to see a unit.
- Contractor shall coordinate and develop a change package for all comments, questions, and potential changes requested by the Fabrication vendor(s).
 - This includes any requested design changes of the MLV chassis to be implemented during the fabrication phase.
 - The change package shall include at a minimum:
 - The Fabrication vendor(s)'s comments, questions, or requested change.
 - The proposed redlines for changes or the proposed response.
 - The Contractor's recommendation with rationale for acceptance or rejection.
 - Previous change board meeting actions for closure.
 - The change package shall be delivered to NASA personnel, including WFF Engineering, BPO Technology Development, and Balloon Program (TM and COR).
 - It shall be delivered no later than five (5) workdays prior to the change board meeting.
- Contractor shall address all comments received from the review cycle.
 - If the comments require a change, the Contractor shall coordinate and incorporate the requested change.
- Contractor shall schedule, deliver, and facilitate a review cycle for
 - All change packages.
 - Updates to the 08-20010-DOC-08 Fabrication Recommendations and Procedures Document, including but not limited to: identification of required OSHA fall arrest/restraint anchor points mounts (ref: 08-20000).
 - Updates to the 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan.
 - Draft NBOC MLV Post Acceptance Test Inspection Plan.
 - MLV Fabrication Inspections and Tests Document.
- Contractor shall schedule change board meetings and present to NASA BPO for approval of the following:
 - All change packages.
 - Updates to the 08-20010-DOC-08 Fabrication Recommendations and Procedures Document.
 - Updates to 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan.
 - Draft NBOC MLV Post Acceptance Test Inspection Plan.
 - The meeting shall include the appropriate Fabrication vendor(s), Contractor, and NASA personnel, including WFF Engineering, TM, and BPO COR. NASA Wallops Safety will be invited as required.

- NASA BPO Program Manager or designee is the acceptance authority for any changes to design.
- Contractor shall update the MLV Design package, as required.
- Contractor shall deliver updated MLV Design package to NASA BPO, including a draft NBOC MLV Post Acceptance Test Inspection Plan and a MLV Fabrication Inspections and Tests Document.
- Contractor shall provide recommendations to NASA on initiating fabrication of the MLV chassis.

For Reference only

SCHEDULE C – MLV FABRICATION

TASK DESCRIPTION

- This schedule is the actual fabrication of the MLV chassis in accordance with the MLV Chassis design package and documents from schedule B.
- Definitions for use in this schedule:
 - **Rework:** Action on a nonconforming product to make it conform to the requirements (e.g. Return to Print, Return to Drawing, Return to Specification).
 - **Repair:** Action on a nonconforming product to make it acceptable for the intended use (e.g. not returned to print).

MATERIALS, EQUIPMENT AND/OR TOOLS

- Government Owned / Government Furnished - Caterpillar 966M wheel loader, if requested.
- Government Furnished Property – MLV Launch Head, if requested.
- Material as noted in drawings.
- Equipment for inspections as noted in drawings.

STANDARDS, SPECIFICATIONS, AND/OR NASA REQUIREMENT DOCUMENTS

- OSHA 29 CFR 1910: General Industry Standards
- State and Federal Regulations, as applicable intended use of MLV chassis
- Draft NBOC MLV Acceptance Test Inspection Plan from schedule B.
- Draft MLV Fabrication Inspections and Tests Document from schedule B.
- See MLV Design Package in the General Requirements (as updated from schedule B), including but not limited to:
 - 08-20010-DOC-08 Fabrication Recommendations and Procedures Document
 - 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan
 - 08-20010-DOC-09 Maintenance Documentation

TASK DELIVERABLES AND MILESTONES

- Deliver a minimum of one (1), a maximum of three (3) MLV Chassis to NASA
 - One (1) MLV Chassis **no later than November 2026.**
- Schedule of deliverables for the MLV Chassis
 - Electronic format
- Successful training of CSBF personnel for operations of the MLV chassis.
- Successful training of CSBF personnel for operations of the fully assembled MLV.
- As-built drawings for the MLV chassis.
 - Electronic format in Adobe PDF format and associated STEP files
- Updated 08-20010-DOC-09 Maintenance Documentation.
 - Electronic format
- An updated 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan
 - Electronic format
- NBOC MLV Post Acceptance Test Inspection Plan
 - Electronic format
- NBOC MLV Post Acceptance Test Inspection Report.
 - Electronic format

- Updated MLV Fabrication Inspections and Tests Document.
 - Electronic format
- NBOC MLV Acceptance Test Repairs and Rework Document.
 - Electronic format
- MLV Operations Document
 - Electronic format
- Bi-weekly high level status summary
 - Electronic format

REQUIREMENTS

- Contractor shall fabricate a minimum of one (1), with the option to deliver a maximum of three (3) MLV Chassis with intended delivery to NASA.
 - One (1) MLV Chassis shall be delivered no later than November 2026.
- Contractor shall provide a single point of contact (POC) to serve as the primary technical interface for the MLV project to NASA Task Manager (TM).
 - The Contractor POC shall schedule meetings with NASA, as needed, to provide the progress and status of the project.
 - Meeting attendance shall be by telecon or on-line meeting platform.
 - Contractor POC shall provide bi-weekly high-level status summary to NASA TM and NASA BPO COR.
 - The high-level status summary shall include all support elements of the SOW.
- Contractor shall provide a schedule for all activities outlined within this schedule.
 - The schedule shall include, at a minimum, the following:
 - Timeline for each deliverable.
 - Tentative dates for Reviews, Testing, and Inspections.
 - If multiple vendors, it shall be a fully integrated schedule with each vendor's timeline denoted.
 - The schedule shall be delivered within two (2) weeks of award to Fabrication vendor(s).
- Contractor shall provide consultation services for to the Fabrication vendor(s) for:
 - Fabrication of the MLV Chassis.
 - Potential design changes of the MLV chassis requested during the fabrication phase.
 - The intended operational implementation of the MLV chassis.
 - End-use operational environments.
 - The support of updating trade studies associated with the maintenance and life cycle considerations of the MLV chassis.
 - Updates of 08-20010-DOC-09 Maintenance Documentation.
 - Development of MLV Operations document. This document shall include:
 - End-use operational environments;
 - Intended operations of the MLV chassis; and
 - Intended operations of the fully integrated MLV.
 - Necessary training and/or license(s) to operate fully integrated MLV.
 - Intended operations for preparing the MLV for transportation.
 - Updates to 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan.
 - This plan is inclusive and representative of all the requirements levied.
 - This plan is a means of verifying the design, requirements, and capabilities of the system.

- This plan shall include both the MLV Chassis and the fully integrated MLV.
- The plan shall include a section indicating role and responsibilities for each of the various testing procedures. (e.g. Vendor X to perform test A; Contractor to perform test C)
- Updates to NBOC MLV Post Acceptance Test Inspection Plan.
 - The plan shall:
 - Ensure the operational integrity of the MLV chassis and the fully assembled MLV;
 - Identify the procedures for each test outlined in 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan;
 - Identify what will be inspected on the MLV chassis to ensure its operational integrity;
 - Identify what will be inspected on the fully assembled MLV to ensure its operational integrity; and
 - Identify what will be documented in the NBOC MLV Post Acceptance Test Inspection Report, including but not limited to the results of:
 - tests in 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan; and
 - inspections detailed in NBOC MLV Acceptance Test Inspection Plan.
- Updates to the MLV Fabrication Inspections and Tests Document
 - The document shall:
 - Identify inspections and/or tests that will be performed during fabrication.
 - Identify any certifications required to perform the inspections and/or tests.
 - Be a repository for the documentation of inspections and the results.
 - Be a repository for the documentation of tests and the results.
- Contractor shall provide any other reasonable support required for the successful fabrication of the MLV chassis.
 - Support may include, but is not limited to:
 - Facilitating access to a Caterpillar 966M in the most cost-effective method possible. This may be through leasing a unit for them to host at their facility, transporting a unit that is currently in the Contractor's possession to the Fabrication vendor, and/or hosting the Fabrication vendor at the CSBF facility to see a unit.
 - As requested, facilitating access to a MLV Launch head in the most cost-effective method possible.
- Contractor shall coordinate and develop a change package for:
 - All comments, questions, and potential changes requested by the Fabrication vendor;
 - Any requested changes of the MLV chassis design by the Fabrication vendor;
 - Any change to drawings, documentation, or requirements;
 - Any repair;
 - The change package shall include at a minimum:
 - The current requirement (e.g. drawing, document, specification, requirement);
 - The proposed changes with the rationale for the request;
 - The Contractor's recommendation with rationale for acceptance or rejection.
 - Be configuration management controlled.

- If the change package is for a repair, then it shall include at a minimum:
 - The current requirement (e.g. drawing, document, specification)
 - The proposed detailed steps to perform the repair;
 - Provide rationale as to why it cannot be reworked;
 - Contractor's recommendation with rationale for acceptance or rejection;
 - Be configuration management controlled.
- The change package format shall be:
 - Microsoft Office and/or Adobe compatible format.
 - Drawing change packages shall be in Adobe PDF format with associated STEP files.
- Contractor shall schedule, deliver, and facilitate a review cycle for:
 - All change packages, including but not limited to:
 - All nonconformances not returned to print.
 - Updates to the 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan
 - Updates to 08-20010-DOC-09 Maintenance Documentation.
 - Updates NBOC MLV Acceptance Test Inspection Plan.
 - Updates to MLV Fabrication Inspections and Tests Document.
 - MLV Acceptance Test Repairs and Rework Document.
 - MLV Operations Document
 - NBOC MLV Post Acceptance Test Inspection Report
 - Updates to the MLV Design Package drawings
 - All change packages shall be delivered to the change board members no later than five (5) workdays prior to the change board meeting.
 - The review shall include the appropriate Fabrication vendor, NASA and Contractor personnel, including Engineering, Technology Development, and Balloon Program Office (Task Manager and COR).
 - Any change package affecting the NASA owned MLV Design package shall be accepted by NASA prior to implementation.
- Contractor shall address all comments received from the review cycle.
 - If the comments require a change, the Contractor shall coordinate and incorporate the requested change.
 - The Fabrication Vendor may be consulted for disposition of comments.
- Contractor shall schedule change board meetings and present for approval of the following:
 - All change packages.
 - Final review of 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan.
 - This document shall be approved by NASA prior completion of fabrication of MLV Chassis.
 - Final review of NBOC MLV Acceptance Test Inspection Plan.
 - This document shall be approved by NASA prior to acceptance testing of the MLV Chassis.
 - Final review of NBOC MLV Post Acceptance Test Inspection Report
 - This document shall be approved by NASA prior to acceptance of the completed MLV chassis fabrication.
 - Final review of MLV Acceptance Test Repairs and Rework Document
 - This document shall be approved by NASA prior to acceptance of the completed MLV chassis fabrication.

- The meeting shall include the appropriate Fabrication vendor, Contractor, and NASA personnel, including Engineering, TM, and BPO COR. NASA Wallops Safety will be invited as required.
 - NASA BPO Program Manager or designee is the acceptance authority.
- After fabrication of MLV Chassis is complete, Contractor shall:
 - Implement the 08-20010-DOC-05 NBOC MLV Acceptance Testing Plan.
 - This may be through the supervision of the Fabrication Vendor, a third party, and/or through Contractor in-house capabilities.
 - Document all deficiencies and/or discrepancies found in the NBOC MLV Acceptance Test Inspection Report
 - Perform the NBOC MLV Acceptance Test Inspection Plan.
 - Document all deficiencies and/or discrepancies in the NBOC MLV Acceptance Test Inspection Report.
 - Deliver the NBOC MLV Acceptance Test Inspection Report.
- Contractor shall develop and deliver the MLV Acceptance Test Repairs and Rework Document to address the NBOC MLV Acceptance Test Inspection Report.
 - The document shall be the repository for all repairs and rework.
 - The document shall:
 - Include photographic evidence of rework and/or repair (before and after).
 - All the repair(s) shall be submitted through the change package process and approved by NASA prior to implementation.
- Contractor shall provide a complete MLV As-Built drawing set of the MLV Chassis
 - The as-built drawings shall reflect any repairs or rework, including those resulting from acceptance testing.
 - The as-built drawings format shall be in Abode PDF format.
 - The as-built drawings format shall also include the associated STEP file.
- Contractor shall ensure the appropriate personnel are fully trained in the
 - Operations of the MLV Chassis;
 - Operations of the fully assembled and integrated MLV;
 - Maintenance of the MLV Chassis
- Contractor shall provide recommendations to NASA on the acceptance of the MLV chassis.
- Contractor shall deliver a minimum of one (1), with the option to deliver a maximum of three (3) MLV Chassis to NASA.
 - NASA has the right to inspect and accept the MLV Chassis with similar procedures to FAR 52.246-3, Inspection of Supplies – Cost-Reimbursement (May 2001).
 - The MLV Chassis Design is owned by NASA BPO.
 - One (1) fully acceptable MLV Chassis shall be delivered **no later than November 2026** to NASA.